

Departement Elektriese, Elektroniese en
Rekenaar-Ingenieurswese

Department of Electrical, Electronic and
Computer Engineering

Semestertoets 1
VRAESTEL

Kopiereg voorbehou

Semester Test 1
QUESTION PAPER

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Module EBN111
Elektrisiteit en Elektronika
Maart 2017

Module EBN111
Electricity and Electronics
March 2017



Memorandum

UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Eksamensinligting:

Exam information:

Maksimum punte: <i>Maximum marks:</i>	46	Volpunte: <i>Full marks:</i>	45
Duur van vraestel: <i>Duration of paper:</i>	90 min + 10 min to complete OCR form <i>90 min + 10 min om OKH vorm te voltooi</i>	Oopboek / toeboek: <i>Open / closed book:</i>	Toe <i>Closed</i>
Totale aantal bladsye (hierdie blad ingesluit): <i>Total number of pages (including this page):</i>		15 + 2 Answer sheets	

BELANGRIK- IMPORTANT

- Die eksamenregulasies van die Universiteit van Pretoria geld.
The examination regulations of the University of Pretoria apply.
- Alle vrae moet op die ANTWOORDBLAD ingevul word, en ook op die Optiese Karakter Herkenning (OKH) vorm.
All questions must be answered on the ANSWER SHEET and on the Optical Character Recognition (OCR) form.
- Neem noukeurig kennis van die instruksies op die OKH vorm.
Take careful consideration of the instructions on the OCR form.
- Antwoorde moet eenhede insluit!
Answers must include units!
- Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.
Final answers must be written as real numbers, and not as fractions.

Dosent(e): <i>Lecturer(s):</i>	1. D le Roux
	2. D Ramotsoela
	3. X Ye

SECTION / AFDELING 1 [18]

Question 1 & 2 [2]		Vraag 1 & 2 [2]	
01	An electric stove with two electric plates and an oven is used in preparing a meal as follows: • Plate 1 (1.8 kW) : 20 minutes, • Plate 2 (1.2 kW): 40 minutes, • Oven (3 kW): 30 minutes Calculate the cost of electricity used in preparing the meal. (Electricity costs R 2 per kWh. Give your answer in Rands.) [1]	01	'n Elektriese stoof met twee elektriese plate en een oond word as volg gebruik om 'n maaltyd voor te berei: • Plaat 1 (1.8 kW): 20 minute • Plaat 2 (1.2 kW): 40 minute • Oond (3 kW): 30 minute Wat is die elektrisiteitskoste om die maaltyd voor te berei? (Elektrisiteit kos R 2 per kWh. Gee die antwoorde in Rande.) [1]
02	If a constant 100 V is applied to an element, and the element uses 150 kJ over 1 min, what is the size of the current flowing through the element? [1]	02	Indien 'n konstante 100 V oor 'n element aangelê word, en die element gebruik 150 kJ oor 1 min, wat is die grootte van die stroom deur die element? [1]

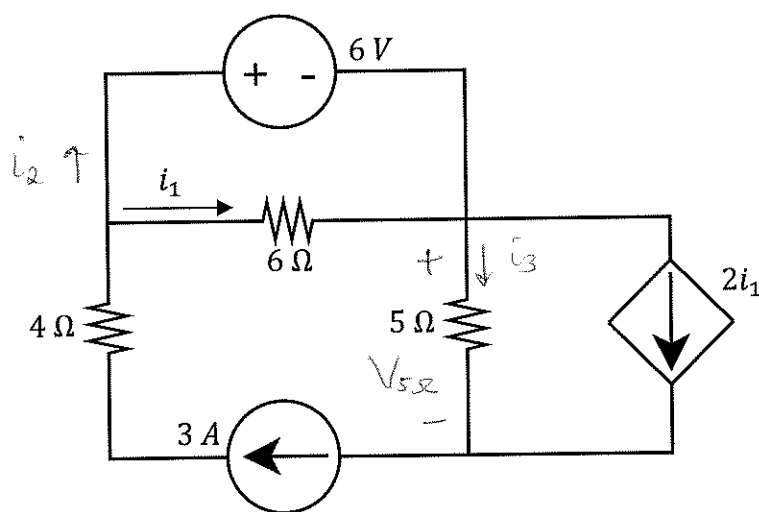
$$\begin{aligned}
 * \text{ Cost} &= (1.8 \text{ kW} \times \frac{20}{60} \text{ h} + 1.2 \text{ kW} \times \frac{40}{60} \text{ h} \\
 &\quad + 3 \text{ kW} \times \frac{30}{60} \text{ h}) \times R_2 / \text{kWh} \\
 &= \underline{R 5.80} \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 * \quad V_s &= 100 \text{ V} \\
 W &= 150 \text{ kJ} \\
 t &= 60 \text{ s.}
 \end{aligned}$$

$$\begin{aligned}
 P &= \frac{\Delta W}{\Delta T} = \frac{150 \text{ kJ}}{60 \text{ s}} \\
 &= 2.5 \text{ kW}
 \end{aligned}$$

$$\begin{aligned}
 \text{but } P &= V_s \cdot I \\
 \therefore I &= \frac{2.5 \text{ kW}}{100} \\
 &= \underline{25 \text{ A}} \quad \checkmark
 \end{aligned}$$

Question 3 & 4 [4]		Vraag 3 & 4 [4]	
Find the power related to following elements in the circuit below:		Bepaal die drywing geassosieer met die volgende elemente in onderstaande kringbaan:	
03	The independent voltage source.	03	Die onafhanklike spanningsbron.
04	The dependent current source.	04	Die afhanklike stroombron.



$$* \quad i_1 = \frac{6}{6} = 1A.$$

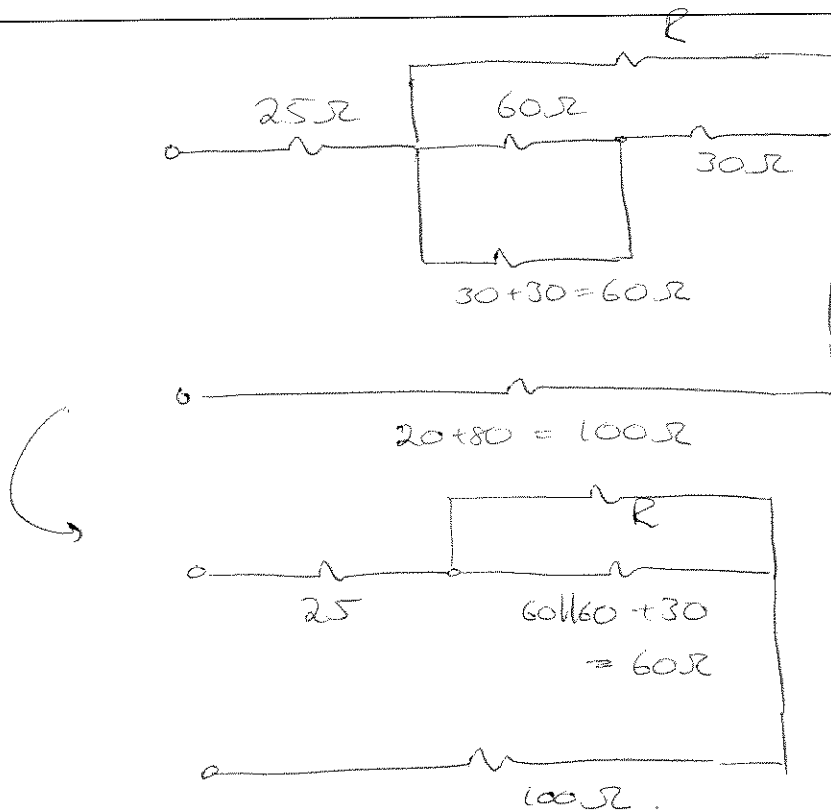
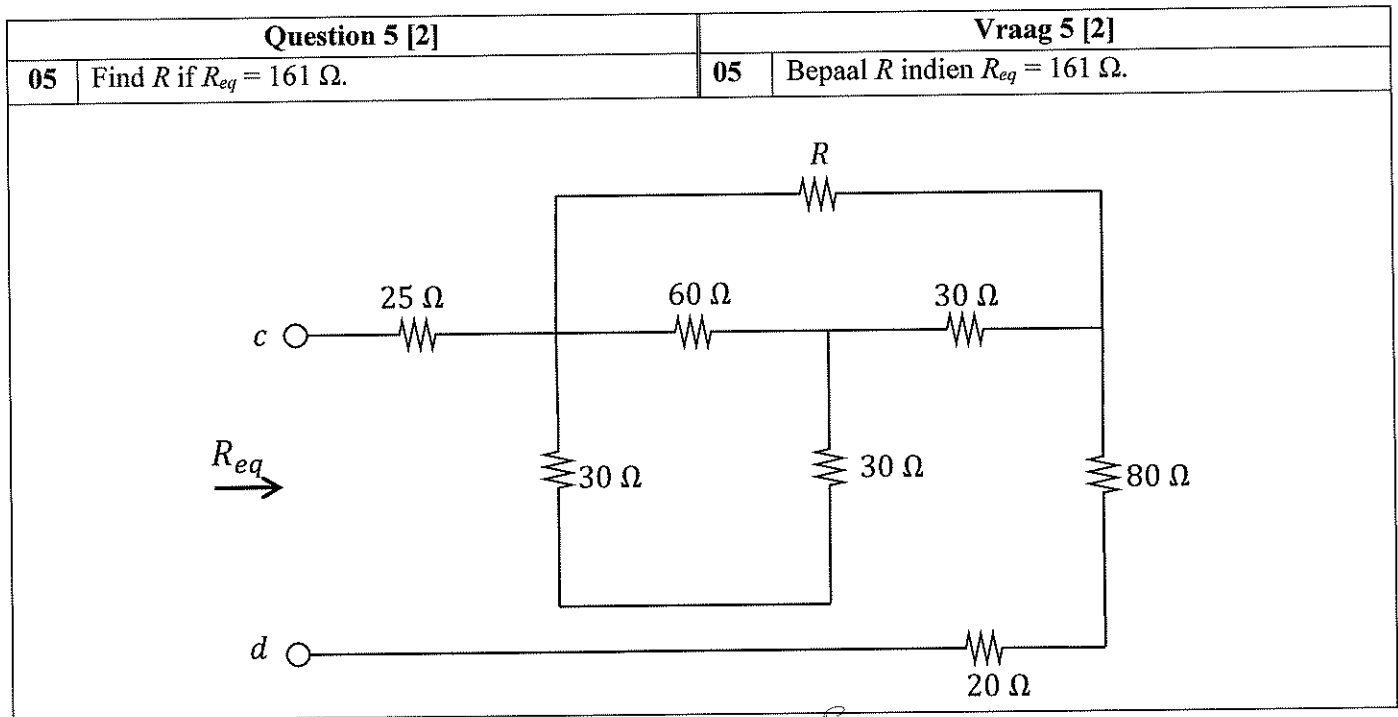
$$\rightarrow \text{KCL: } i_2 = 3 - i_1 = \underline{2A}$$

$$\rightarrow P_{6V} = +6 \times 2 = \underline{12W} \quad \checkmark$$

$$* \text{ KCL: } i_3 = 3 - 2i_1 = 1A.$$

$$\rightarrow \text{KVL: } V_{5\Omega} = +5 \times 1 = 5V.$$

$$\therefore P_{\diamond} = +5 \times (2i_1) \\ = \underline{10W} \quad \checkmark$$



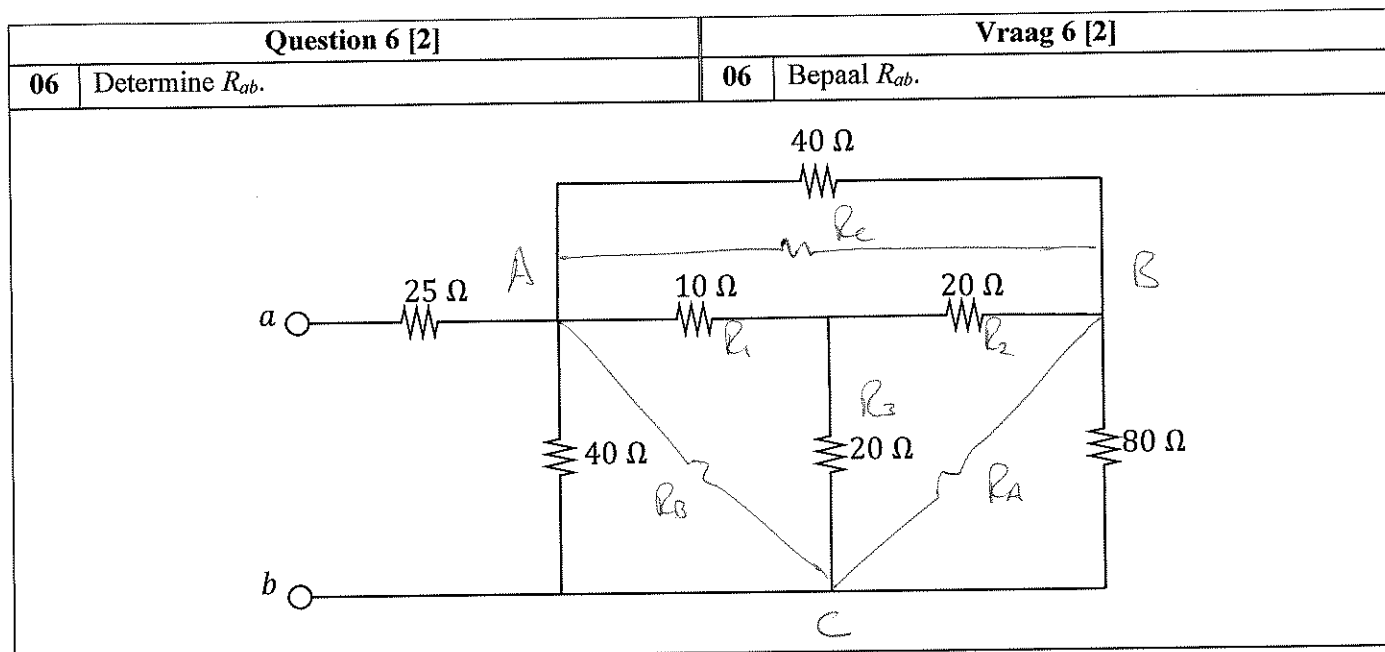
$$R_{eq} = 161 = 25 + R \parallel 60 + 100$$

$$R \parallel 60 = 36$$

$$\frac{R \times 60}{R + 60} = 36$$

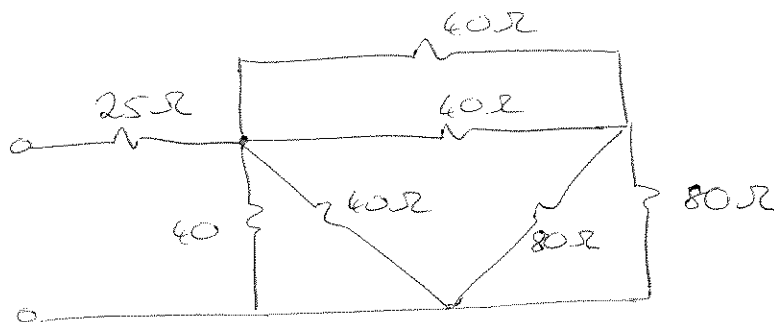
$$60R = 36R + 2160$$

$$R = \frac{2160}{60 - 36} = 90 \Omega$$



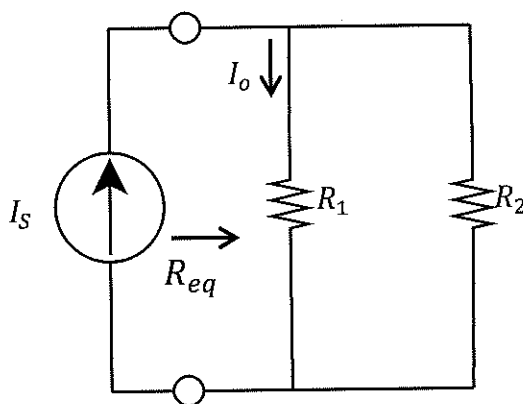
$$\begin{aligned}
 * R_A &= \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1} \\
 &= \frac{10 \times 20 + 20 \times 20 + 20 \times 10}{10} \\
 &= \frac{800}{10} = 80 \Omega
 \end{aligned}$$

$$* R_B = \frac{800}{20} = 40 \Omega = R_C$$



$$\begin{aligned}
 R_{ab} &= 25 + (40 \parallel 40) \parallel (40 \parallel 40 + 80 \parallel 80) \\
 &= 25 + 20 \parallel (20 + 40) \\
 &= 25 + \frac{20 \times 60}{80} \\
 &= \underline{40 \Omega} \quad \checkmark
 \end{aligned}$$

Question 7 & 8 [4]				Vraag 7 & 8 [4]			
For the circuit below it is given that $\frac{I_o}{I_s} = 0.2$ and $R_{eq} = 5 \text{ k}\Omega$. Find R_1 and R_2 .				Vir onderstaande kringbaan is dit gegee dat $\frac{I_o}{I_s} = 0.2$ en $R_{eq} = 5 \text{ k}\Omega$. Bepaal R_1 en R_2 .			
07	R_1	08	R_2	07	R_1	08	R_2



$$I_o = 0.2 I_s$$

$$\text{but } I_o = \frac{R_2 I_s}{R_1 + R_2} \quad (\text{Current division})$$

$$\therefore 0.2 = \frac{R_2}{R_1 + R_2}$$

$$0.2 R_1 = R_2 - 0.2 R_2$$

$$\underline{R_1 = 4 R_2}$$

$$* R_{eq} = 5k = \frac{R_1 R_2}{R_1 + R_2}$$

$$5k R_1 + 5k R_2 = R_1 R_2$$

$$\text{Substitute: } 5k(4R_2) + 5k R_2 = R_2(4R_2)$$

$$0 = 4 R_2^2 - 25k R_2$$

$$\therefore R_2 = \frac{25k}{4} = \underline{6.25k\Omega} \quad \checkmark$$

$$R_1 = \underline{25k\Omega} \quad \checkmark$$

Question 9 & 10 [4]				Vraag 9 & 10 [4]			
If $I_0 = 5$ A, determine I_S and V_1 .				Indien $I_0 = 5$ A, bepaal I_S en V_1 .			
09	I_S	10	V_1	09	I_S	10	V_1

$$* I_0 = \frac{-I_S \times 50}{30 + 50} \quad (\text{Current division})$$

$$5(80) = -I_S \times 50$$

$$I_S = \underline{-8\text{ A}} \quad \checkmark \checkmark$$

$$* V_1 = \frac{(5I_0)}{30 + 50} \times 50 \quad (\text{Volt division})$$

$$= \frac{25}{80} \times 50$$

$$= \underline{15.625\text{ V}} \quad \checkmark \checkmark$$

VRAAG/QUESTION 2 [20]

Question 11 & 12 [4]		Vraag 11 & 12 [4]	
11	Determine θ if $v_1 = -2$ V. $\begin{bmatrix} \theta & -2 \\ 6 & 4 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} \theta \\ -3\theta \end{bmatrix}$	11	Bepaal θ indien $v_1 = -2$ V. $\begin{bmatrix} \theta & -2 \\ 6 & 4 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} \theta \\ -3\theta \end{bmatrix}$
12	Determine σ if $i_2 = -2.6$ A. $\begin{bmatrix} 2 & 1 & 4 \\ 0 & \sigma & 1 \\ 2 & 4 & 0 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ -2 \end{bmatrix}$	12	Bepaal σ indien $i_2 = -2.6$ A. $\begin{bmatrix} 2 & 1 & 4 \\ 0 & \sigma & 1 \\ 2 & 4 & 0 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ -2 \end{bmatrix}$

$$\Delta = \begin{vmatrix} \theta & -2 \\ 6 & 4 \end{vmatrix} = 4\theta + 12.$$

$$\Delta_1 = \begin{vmatrix} \theta & -2 \\ -3\theta & 4 \end{vmatrix} = 4\theta - 6\theta = -2\theta$$

$$v_1 = -2 = \frac{\Delta_1}{\Delta} \Rightarrow -2 = \frac{-2\theta}{4\theta + 12}$$

$$4\theta + 12 = \theta$$

$$3\theta = -12$$

$$\theta = \underline{-4} \quad \checkmark \checkmark$$

$$\Delta = \begin{vmatrix} 2 & 1 & 4 \\ 0 & \sigma & 1 \\ 2 & 4 & 0 \end{vmatrix} = 2 - 8\sigma - 8$$

$$= -6 - 8\sigma$$

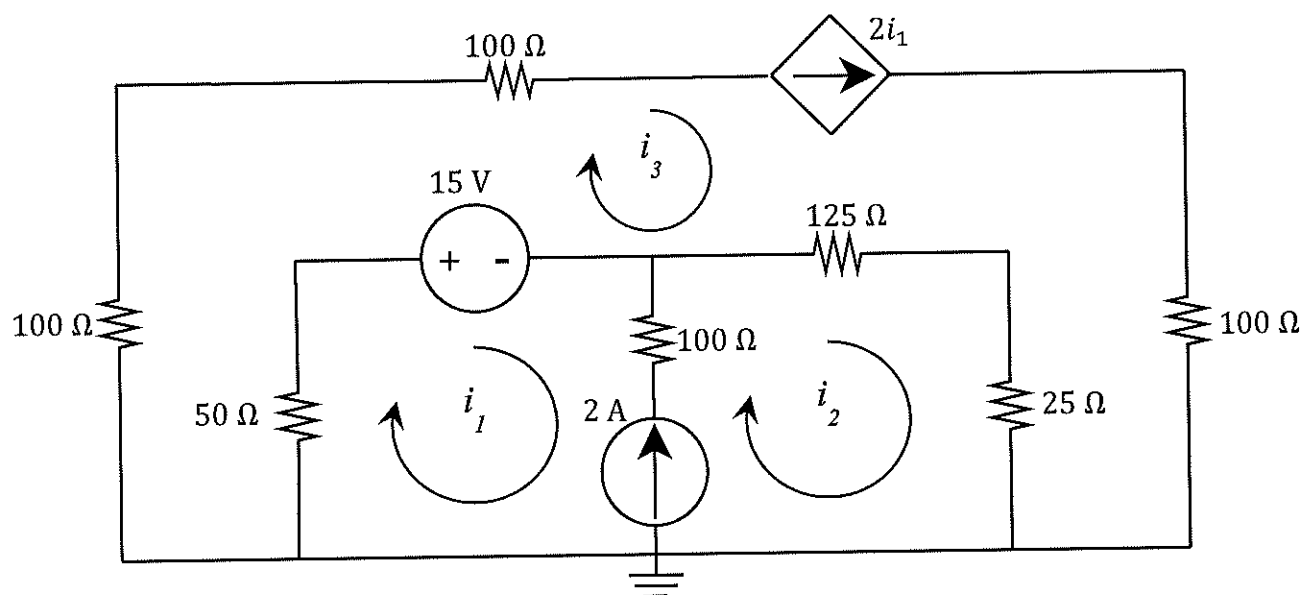
$$\Delta_2 = \begin{vmatrix} 2 & 1 & 4 \\ 0 & \sigma & 1 \\ 2 & -2 & 0 \end{vmatrix} = +2 - 32 + 4 = -26.$$

$$i_2 = -2.6 = \frac{\Delta_2}{\Delta} = \frac{-26}{-6 - 8\sigma}$$

$$6 + 8\sigma = -10$$

$$\sigma = \underline{-2} \quad \checkmark \checkmark$$

Question 13-16 [4]				Vraag 13-16 [4]			
Use mesh analysis to set up two equations of the following form: $ai_1 + bi_2 = -15$ $ci_1 + di_2 = 2$ Calculate the unknown coefficients.				Gebruik lusanalyse om twee vergelykings van die volgende vorm op te stel: $ai_1 + bi_2 = -15$ $ci_1 + di_2 = 2$ Bepaal die onbekende koëffisiënte.			
13	a	14	b	13	a	14	b
15	c	16	d	15	c	16	d



$$i_3 = 2i_1$$

$$\text{KCL} \quad 2 = i_2 - i_1 \rightarrow -i_1 + i_2 = 2$$

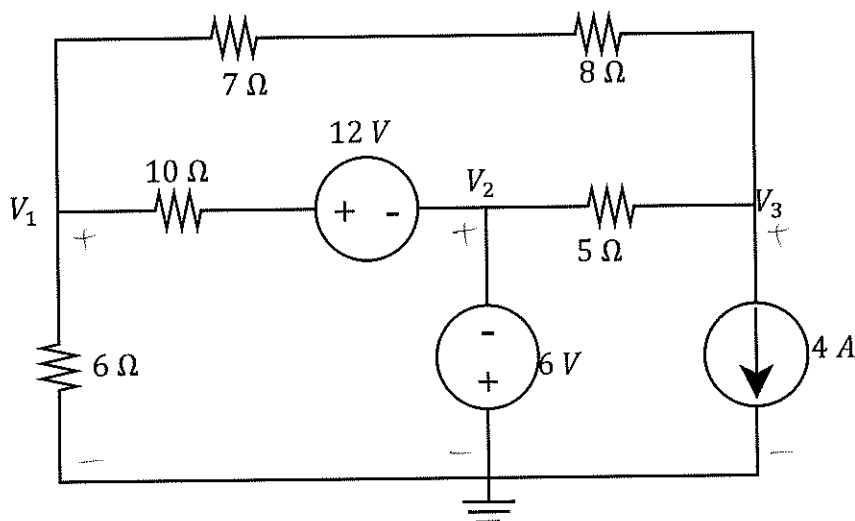
$$\text{Supermesh} \quad 50i_1 - 50i_3 + 15 + 125i_2 - 125i_3 + 25i_2 - 25i_3 = 0$$

$$50i_1 + 150i_2 - 20i_3 = -15$$

$$\rightarrow \text{but } i_3 = 2i_1$$

$$\therefore -350i_1 + 150i_2 = -15$$

Question 17-20 [4]				Vraag 17-20 [4]			
Use nodal analysis to set up two equations of the following form: $eV_1 + fV_3 = 18$ $gV_1 + hV_3 = -78$ Calculate the unknown coefficients.				Gebruik nodeanalyse om twee vergelykings in die volgende vorm op te stel: $eV_1 + fV_3 = 18$ $gV_1 + hV_3 = -78$ Bepaal die onbekende koëffisiënte.			
17	e	18	f	17	e	18	f
19	g	20	h	19	g	20	h



$$V_2 = -6V$$

$$\textcircled{1} \quad \frac{V_1}{6} + \frac{V_1 - 12 - V_2}{10} + \frac{V_1 - V_3}{15} = 0$$

$$(\times 30) \quad 5V_1 + 3V_1 - 3(12 - 6) + 2V_1 - 2V_3 = 0$$

$$10V_1 - 2V_3 = 18$$

$$\textcircled{2} \quad 4 + \frac{V_3 - (-6)}{5} + \frac{V_3 - V_1}{15} = 0$$

$$(\times 15) \quad 3V_3 + 18 + V_3 - V_1 = -60$$

$$-V_1 + 4V_3 = -78$$

Question 2.4 [2]

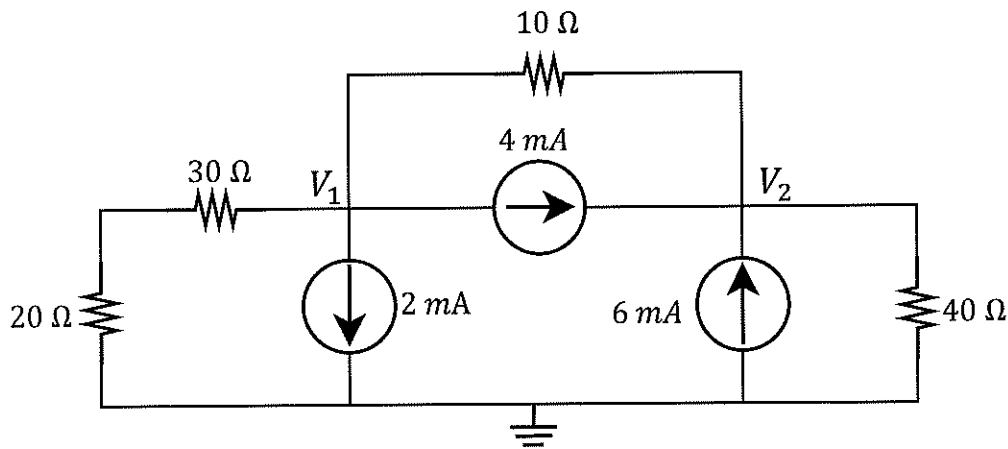
Use inspection to set up the matrix equation in terms of the node voltages.

$$\begin{bmatrix} k & l \\ -0.1 & \rho \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} -6 \text{ mA} \\ q \end{bmatrix}$$

Vraag 2.4 [2]

Gebruik inspeksie om die matriksvergelykings in terme van die nodespannings op te stel.

$$\begin{bmatrix} k & l \\ -0.1 & \rho \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} -6 \text{ mA} \\ q \end{bmatrix}$$



$$\begin{bmatrix} \frac{1}{10} + \frac{1}{30} & -\frac{1}{10} \\ -\frac{1}{10} & \frac{1}{10} + \frac{1}{40} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} (-2 - 4) \text{ mA} \\ (6 + 4) \text{ mA} \end{bmatrix}$$

$$\begin{bmatrix} 0.12 & -0.1 \\ -0.1 & 0.125 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} -6 \text{ mA} \\ 10 \text{ mA} \end{bmatrix}$$

$$k = 0.12$$

$$l = -0.1$$

✓

$$\rho = 0.125$$

$$q = 10 \text{ mA}$$

✓

Question 25 [2]		Vraag 25 [2]	
25	Calculate V_0 using either nodal or mesh analysis.	25	Bepaal V_0 deur gebruik te maak van node- of lus-analise.

$$\frac{V_0}{40} + \frac{V_0 - 10}{10} + \frac{V_0 - 4V_0 + 10}{20} = 1.$$

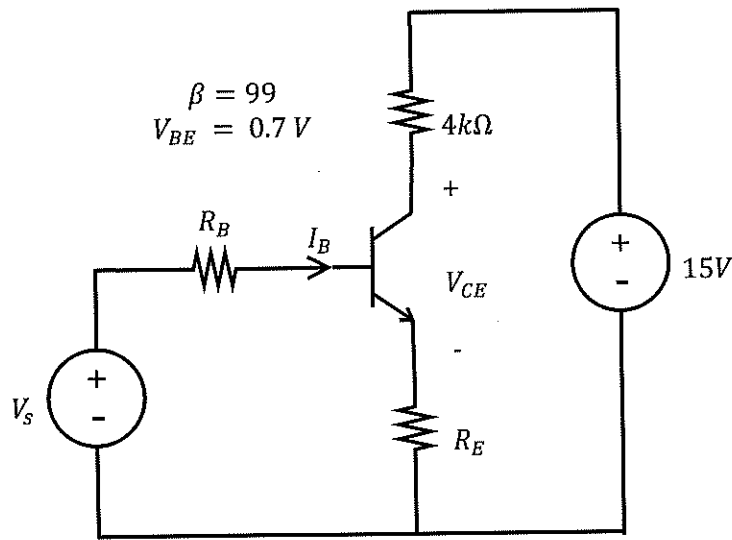
$$(x40) \quad V_0 + 4V_0 - 40 + 2V_0 - 8V_0 + 20 = 40$$

$$-V_0 = 60$$

$$V_0 = -60V$$

✓✓

Question 26 & 27 [4]		Vraag 26 & 27 [4]	
It is given that $\beta = 99$ and $V_{BE} = 0.7V$.		Dit is gegee dat $\beta = 99$ en $V_{BE} = 0.7V$.	
26	Determine V_{CE} if $I_B = 25 \mu A$, $R_E = 0 \Omega$.	26	Determine V_{CE} if $I_B = 25 \mu A$, $R_E = 0 \Omega$.
27	Determine R_B if $I_B = 20 \mu A$, $V_S = 2V$, $R_E = 150 \Omega$.	27	Determine R_B if $I_B = 20 \mu A$, $V_S = 2V$, $R_E = 150 \Omega$.



(26) $-15 + 4k\Omega + V_{CE} + 0 \cdot I_E = 0$

$$V_{CE} = 15 - 4k\Omega (99)(25\mu)$$

$$= \underline{5.1V} \quad \checkmark$$

(27) $-2 + 20\mu \cdot R_B + 0.7 + 150 \times I_E = 0.$

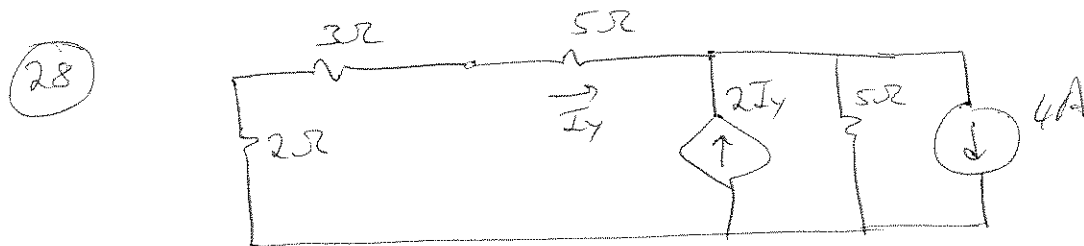
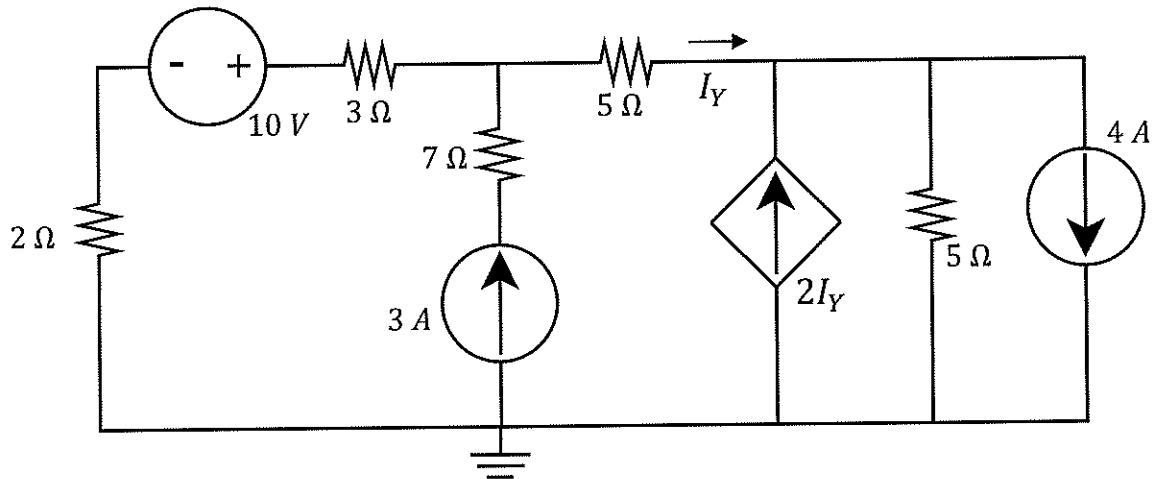
$$20\mu R_B = 1.3 - 150(99+1)(20\mu)$$

$$R_B = \frac{1}{20\mu}$$

$$= \underline{50k\Omega} \quad \checkmark$$

VRAAG/QUESTION 3 [8]

Question 28 & 29 [4]		Vraag 28 & 29 [4]	
Using superposition and calculate:		Gebruik superposisie en bepaal:	
28	the current I_Y due to the 4 A current source alone.	28	die stroom I_Y slegs as gevolg van die 4 A stroombron.
29	the current I_Y due to the 10 V voltage source alone.	29	die stroom I_Y slegs as gevolg van die 10 V spanningsbron.

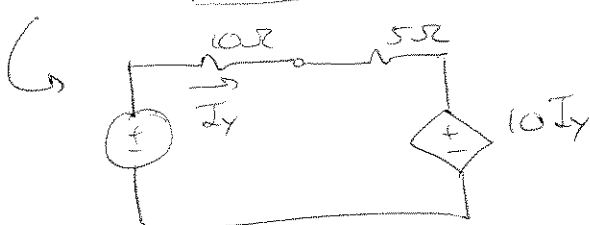
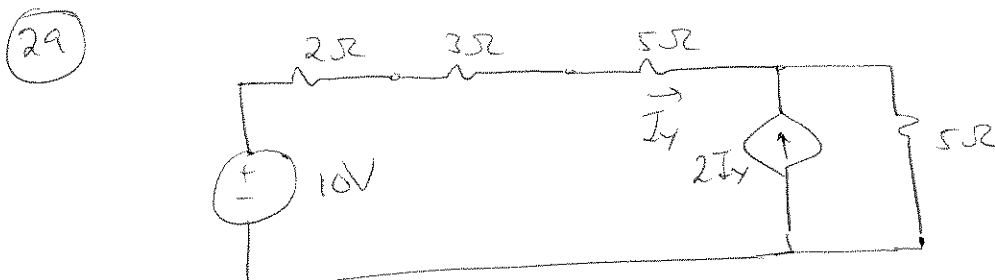


$$\frac{V_1}{10} + \frac{V_1}{5} + 4 = 2I_Y$$

$$\text{but } I_Y = -\frac{V_1}{10}$$

$$\therefore -I_Y - 2I_Y + 4 = 2I_Y$$

$$I_Y = \frac{4}{5} = 0.8 \text{ A} = \underline{800 \text{ mA}}$$



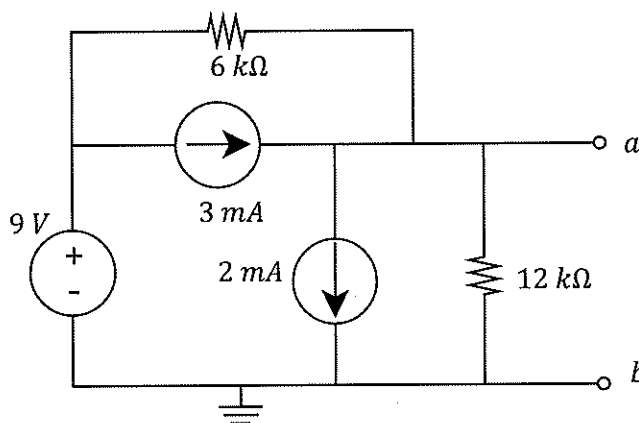
$$I_Y = \frac{10 - 10I_Y}{15}$$

$$15I_Y = 10 - 10I_Y$$

$$I_Y = \frac{10}{25} = \underline{400 \text{ mA}}$$

Question 30 & 31 [4]				Vraag 30 & 31 [4]			
Determine the Thevenin equivalent voltage V_{TH} and resistance R_{TH} w.r.t. terminals $a-b$.				Bepaal die Thevenin ekwivalente spanning V_{TH} en weerstand R_{TH} in terme van terminale $a-b$.			
30	V_{TH}	31	R_{TH}	30	V_{TH}	31	R_{TH}

The circuit diagram shows a 9V DC voltage source with its positive terminal at the top. A 6 kΩ resistor is connected in series with the positive terminal. Following the resistor, the circuit splits into two parallel branches. The first branch contains a 3 mA current source pointing to the right. The second branch contains a 2 mA current source pointing downwards. These two branches recombine, and then a 12 kΩ resistor is connected in parallel with the combined branches. The output terminals are labeled 'a' and 'b', where terminal 'b' is connected to the common ground reference.



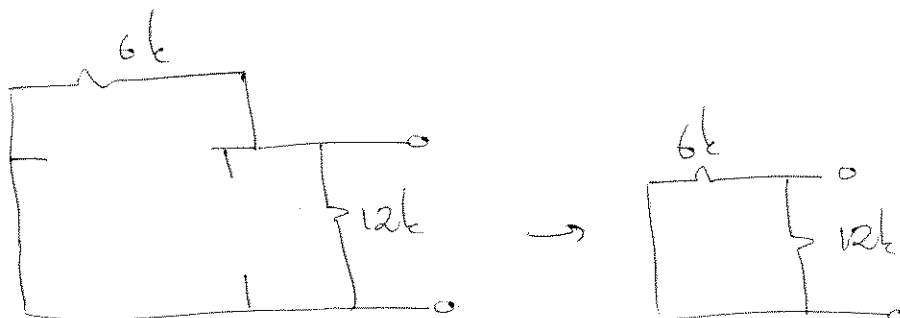
(30)
$$\frac{V_{ab}}{12k} + 2m + \frac{V_{ab} - 9}{6k} - 3m = 0$$

($\times 12k$)
$$V_{ab} + 24 + 2V_{ab} - 18 - 36 = 0.$$

$$3V_{ab} = +30$$

$$V_{ab} = \underline{10V} \quad \checkmark \checkmark$$

(31)



$$R_{TH} = 12k \parallel 6k$$

$$= \frac{12 \times 6}{12 + 6} = \underline{4k\Omega} \quad \checkmark \checkmark$$